SSP

Computer & Communication Department

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Modern Physics
Sheet 6
Solutions

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1. Calculate the momentum of a proton and an electron moving with speed of 0.90c. Convert the answer to MeV/c.

Solution

$$\gamma = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}} = 2.29$$

Proton

$$p = \gamma m u = 2.29 \times 1.67 \times 10^{-27} \times 0.90 \times 3 \times 10^{8} = 1.03 \times 10^{-18} kg.m/s$$

$$= 1.03 \times 10^{-18} \times \frac{3 \times 10^{8}}{10^{6} \times 1.6 \times 10^{-19}} = 1930\ MeV/c$$

Electron

$$p = \gamma m u = 2.29 \times 9.11 \times 10^{-31} \times 0.90 \times 3 \times 10^{8} = 5.64 \times 10^{-22} kg.m/s$$

$$= 5.64 \times 10^{-22} \times \frac{3 \times 10^{8}}{10^{6} \times 1.6 \times 10^{-19}} = 1.06\ MeV/c$$

2. Consider the relativistic form of Newton's second law. Show that when F is parallel to v,

$$F = \gamma^3 ma$$

where *m* is the mass of an object and *a* is its acceleration.

Solution

$$F = \frac{dp}{dt} = \frac{d}{dt}(\gamma mu) = \frac{d}{dt}\left(\frac{mu}{\sqrt{1-\frac{u^2}{c^2}}}\right) = m\frac{d}{dt}\left(u\left(1-\frac{u^2}{c^2}\right)^{-\frac{1}{2}}\right)$$

$$= m\left[\frac{du}{dt}\left(1-\frac{u^2}{c^2}\right)^{-\frac{1}{2}} + u\left(-\frac{1}{2}\right)\left(1-\frac{u^2}{c^2}\right)^{-\frac{3}{2}}\left(-\frac{2u}{c^2}\right)\frac{du}{dt}\right]$$

$$= m\left[\frac{1}{\sqrt{1-\frac{u^2}{c^2}}} + \frac{\frac{u^2}{c^2}}{\left(1-\frac{u^2}{c^2}\right)^{\frac{3}{2}}}\right]\frac{du}{dt} = m\left(1-\frac{u^2}{c^2}\right)^{-\frac{3}{2}}\frac{du}{dt} = \gamma^3 ma$$

3. Show that the energy–momentum relationship given by $E^2 = p^2c^2 + (mc^2)^2$ follows from the expressions $E = \gamma mc^2$ and $p = \gamma mu$.

Solution

$E = \gamma mc^2$ (1)

$p = \gamma mu$ (2)

Square (1) and (2), then multiply (2) by c^2, then subtract (2) from (1)

$$E^2 - (pc)^2 = \gamma^2(mc^2)^2 - \gamma^2m^2c^2u^2 = (mc^2)^2\gamma^2\left(1 - \frac{u^2}{c^2}\right) = (mc^2)^2$$

$$E^2 = (pc)^2 + (mc^2)^2$$

4. An electron has a speed u = 0.850c. Find its total energy and kinetic energy in electron volts.

Solution

$$\gamma = \frac{1}{\sqrt{1-\frac{u^2}{c^2}}} = 1.898$$

$$E = \gamma mc^2 = 1.898 \times 9.11 \times 10^{-31} \times (3 \times 10^8)^2 = 1.56 \times 10^{-13} J$$

$$= 0.973\ MeV$$

Or

$$E = 1.898 \times 0.511 = 0.970\ MeV$$

$$K = E - mc^2 = 0.970 - 0.511 = 0.459\ MeV$$

5. The total energy of a proton is three times its rest energy. (a) Find the proton's rest energy in electron volts. (b) With what speed is the proton moving? (c) Determine the kinetic energy of the proton in electron volts. (d) What is the proton's momentum?

Solution

a)

$$E_0 = mc^2 = 1.67 \times 10^{-27} \times (3 \times 10^8)^2 = 1.50 \times 10^{-13} J = 938\ MeV$$

b)

$$E = 3mc^2$$

$$E = \gamma mc^2$$

$$\gamma = 3$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}} = 3$$

$$u = 0.943c$$

c)

$$K = (\gamma - 1)mc^2 = 2 \times 938 = 1876\ MeV$$

d)

$$p = \frac{1}{c}\sqrt{E^2 - (mc^2)^2} = \frac{1}{c}\sqrt{(3 \times 938)^2 - 938^2} = 2650\ MeV/c$$

6. A proton in a high-energy accelerator is given a kinetic energy of 50 GeV. Determine the (a) momentum and (b) speed of the proton.

Solution

$K = 50000\ MeV$

$E = K + mc^2 = 50938\ MeV$

$$p = \frac{1}{c}\sqrt{E^2 - (mc^2)^2} = \frac{1}{c}\sqrt{50938^2 - 938^2} = 5.09 \times 10^4\ MeV/c$$

$$K = (\gamma - 1)mc^2$$

$$5 \times 10^4 = (\gamma - 1) \times 938$$

$$\gamma = 54.3$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}} = 54.3$$

$$u = 0.9998c$$

7. An electron has a speed of 0.75c. Find the speed of a proton that has (a) the same kinetic energy as the electron and (b) the same momentum as the electron.

Solution

a)

For the electron

$$\gamma = \frac{1}{\sqrt{1-\frac{u^2}{c^2}}} = 1.51$$

$$K = (\gamma - 1)mc^2 = (1.51 - 1) \times 0.511 = 0.261\ MeV$$

For the proton

$$K = (\gamma - 1) \times 938 = 0.261\ MeV$$

$$\gamma = 1.000278$$

$$u = 0.0239\ c = 7.1 \times 10^6\ m/s$$

b)

for the electron

$$p = \gamma m u = 1.51 \times 9.11 \times 10^{-31} \times 0.75 \times 3 \times 10^{8}$$

$$= 3.095 \times 10^{-22}\ kg.m/s$$

For the proton

$$p = \gamma m u = 3.095 \times 10^{-22}\ kg.m/s$$

$$3.095 \times 10^{-22} = \frac{1.67 \times 10^{-27} \times u}{\sqrt{1 - \frac{u^2}{(3 \times 10^8)^2}}}$$

Solving for u

$$u = 1.85 \times 10^{5}\ m/s$$

8. A radium isotope decays to a radon isotope, ^{222}Rn, by emitting an α particle (a helium nucleus) according to the decay scheme $^{226}Ra \rightarrow {}^{222}Rn + {}^{4}He$. The masses of the atoms are 226.0254 (Ra), 222.0175 (Rn), and 4.0026 (He). How much energy is released as the result of this decay?

Solution

$$\Delta m = 226.0254 - (222.0175 + 4.0026) = 0.0053 \text{ amu}$$

$$E = \Delta mc^2 = 0.0053 \times 931.5 = 4.9\ MeV$$

9. Calculate the binding energy in MeV per nucleon in the isotope $^{12}C_6$. Note that the mass of this isotope is exactly 12 u, and the masses of the proton and neutron are 1.007276 u and 1.008665 u, respectively.

Solution

$$\Delta m = 6 \times 1.007276 + 6 \times 1.008665 - 12 = 0.095646\ amu$$

$$BE = \Delta mc^2 = 0.095646 \times 931.5 = 89.1\ MeV$$

$$\frac{BE}{nucleon} = \frac{89.1}{12} = 7.42\ MeV$$

10. An electron having kinetic energy K = 1.000 MeV makes a head-on collision with a positron at rest. (A positron is an antimatter particle that has the same mass as the electron but opposite charge.) In the collision the two particles annihilate each other and are replaced by two γ rays of equal energy, each traveling at equal angles θ with the electron's direction of motion. Find the energy E, momentum p, and angle of emission θ of the γ rays.

Solution

Conservation of momentum in y direction

$$0 = p_1 \sin(\theta) - p_2 \sin(\theta)$$

$$p_1 = p_2$$

$$p_1 c = p_2 c$$

$$E_1 = E_2$$

Conservation of energy

$$K + 2mc^2 = E_1 + E_2$$

$$1.000 + 1.022 = 2E_1$$

$$E = 1.011\ MeV$$

$$p = 1.011\ MeV/c$$

Electron momentum

$$p_e = \frac{1}{c}\sqrt{E^2 - (mc^2)^2} = \frac{1}{c}\sqrt{(1 + 0.511)^2 - 0.511^2} = 1.422\ MeV/c$$

Conservation of momentum in the x-direction

$$p_e = 2pcos(\theta)$$

$$1.422 = 2 \times 1.011 \times \cos(\theta)$$

$$\theta = 45.3°$$